# When constant g is not enough

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Recognize when mgh works near a planet's surface and when it stops working well.

**SUCCESS CRITERIA - I can...**

1. Compare altitude with planetary radius.
2. Explain that gravitational field weakens with center distance.
3. Choose an equation that fits the height change.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A cart begins with **15 J** kinetic energy and gains **7 J** net work. **Find final kinetic energy**.

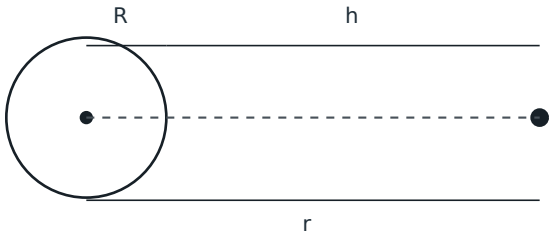
**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $\Delta U \approx mg\,\Delta h$; $g = GM/r^2$.
The first relation needs nearly constant g. For large changes in r, the field changes; one fixed surface value is insufficient.

**Worked reasoning:** A calculation near the surface treats g as constant over small height changes. At a center distance twice a planet's radius, however, g is only one quarter of its surface value.

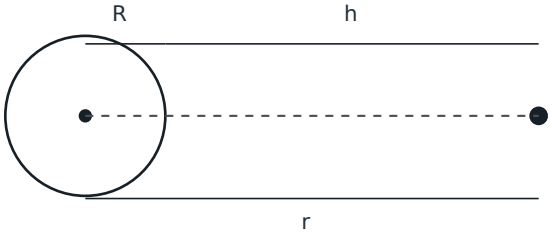
**Your turn:** 1. **Sketch a planet** and **mark center distances R** and **2** R.
2. **Explain why one surface g** cannot describe the entire trip exactly.




**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

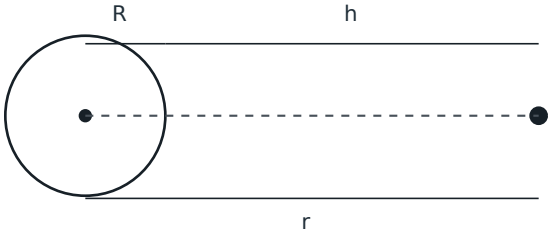
1. **Compare lifting a book 1 m** with moving a satellite one planetary radius above the surface.
2. Which better fits constant g and why?




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Surface g is **12 N/kg**. At center distance **2** R it is **3 N/kg**. Would using **12 N/kg** for the entire trip overestimate or underestimate required lifting energy?
1. Explain.
2. **Label the relevant heights or** distances on the diagram.




**Marzano check - circle:** ① ② ③

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Continue down each column.

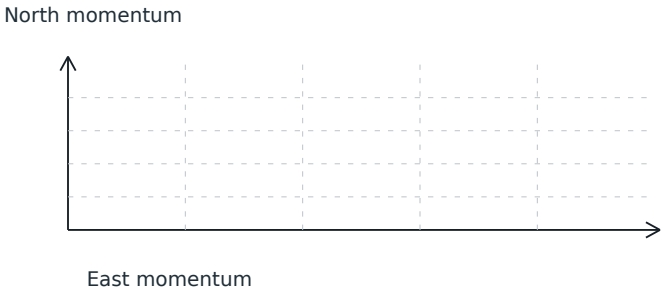
## 05 Explain your reasoning

Why is altitude h not interchangeable with center-to-center r in the universal expression?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

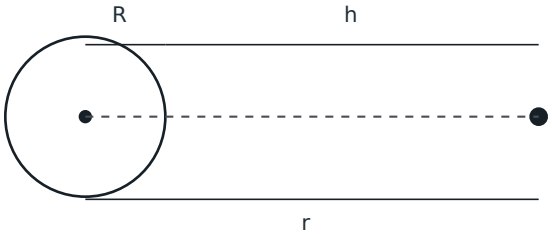
A system's total momentum is **24 kg·m/s** *west* and **32 kg·m/s** *south*. Its combined mass is **4 kg**. **Find final velocity components**.
**Label momentum components** and their sum with units.




**Marzano check - circle:** 1 2 3

## 07 Exit check

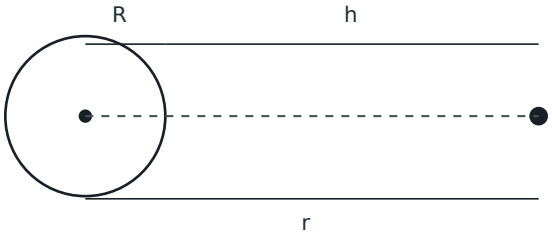
1. **Choose the better description** for a small classroom lift: constant g / strongly changing g.
2. **State a reason**.
3. **Label the relevant heights or** distances on the diagram.




**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A planet's surface field is **12 N/kg**.
1. **Compare it** at altitudes much smaller than R and at h=R.
2. Which is better suited to constant-g reasoning?
3. **Label the relevant heights or** distances on the diagram.




**Marzano check - circle:** 1 2 3